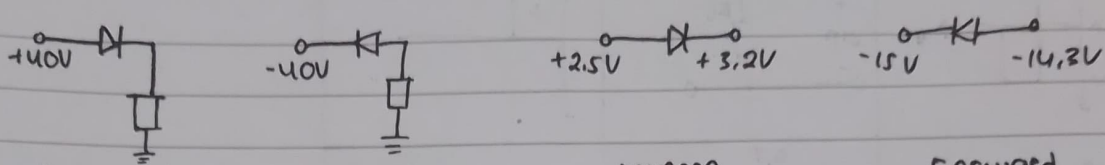
Nama : Ailsa Rahma
NIM : 241331002

No.
Date

Aplikasi Dioda

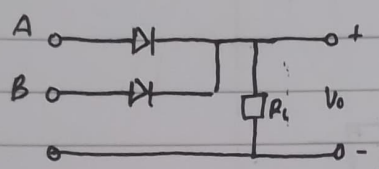
1. Tentukan keadaan masing masing dioda dengan kondisi bias DC



Forward reverse reverse Forword

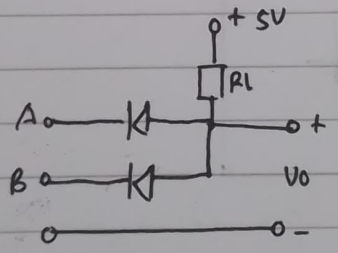
2. Asumsikan dioda ideal. Logika keluaran tiap rangkaian dengan berbagai kemungkinan tegangan masukan.

a).



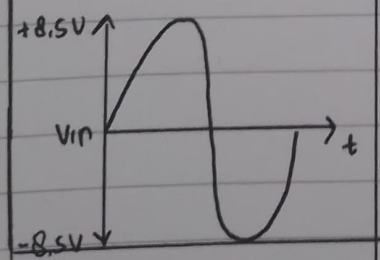
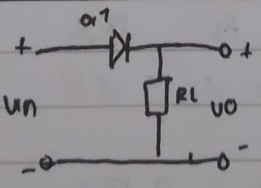
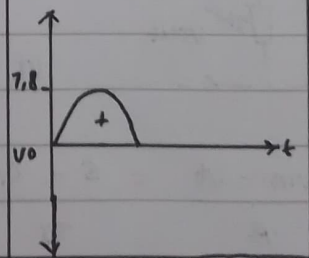
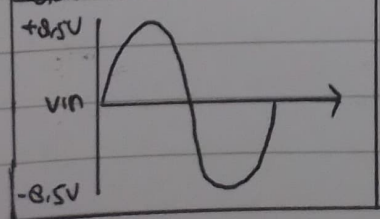
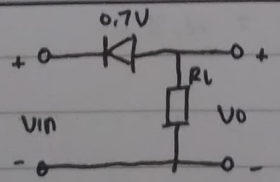
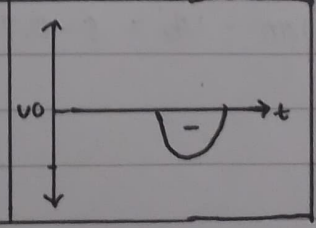
A(V)	B(V)	Vo	Logika
0	0	0	
0	5	5	OR
5	0	5	
5	5	5	

b).

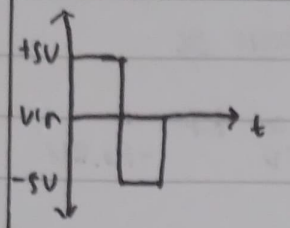
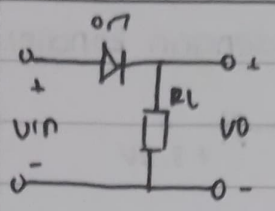
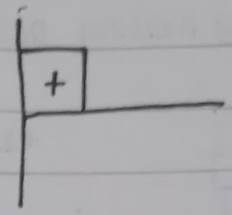
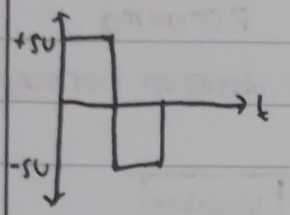
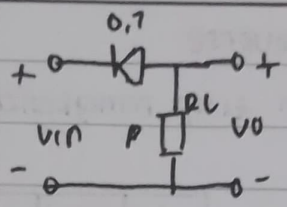
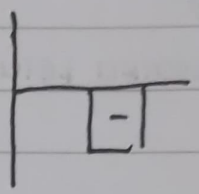
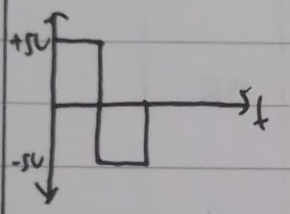
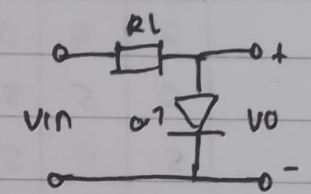
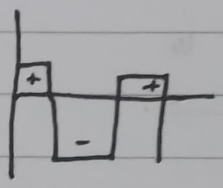
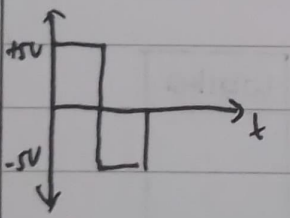
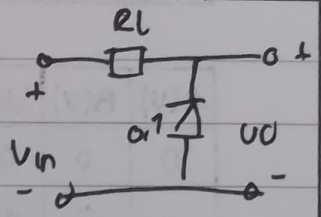
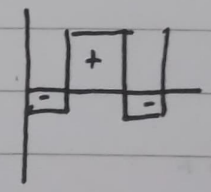


A(V)	B(V)	Vo	Logika
0	0	0	
0	5	0	AND
5	0	0	
5	5	5	

3.

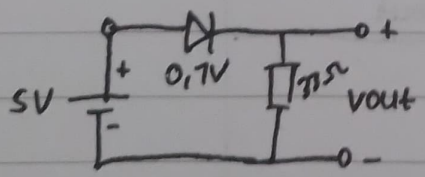
	Vin	Rangkaian Dioda	Vout
a.			
b.			

4. Asumsikan pendekatan dioda dengan V ambang $V_b = 0{,}7V$

Vin	Rangkaian	Vout
		
		
		
		

5. Tentukan besarnya IL dan V pada RL

a). Dioda ideal



$$I_L = \frac{V_S}{R_L} = \frac{5}{33\Omega} = 0{,}15\Omega\Omega\ A$$

$$V_L = V_S = 5V$$

b). $$I_L = \frac{V_{in} - V_b}{R} = \frac{5 - 0{,}7}{33} = 0{,}13A$$

$$V_L = V_{in} - V_b = 5 - 0{,}7 = 4{,}3V$$